PINUX

SMART HOME ARCHITECTURE

The Complete Technical Compendium

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Table of Contents

1. Project Philosophy & Vision 04
2. System Architecture Overview 08
3. Frontend: Apple-Inspired Glassmorphism 14
4. Backend: The Python Asynchronous Brain 22
5. Hardware: ESP32 & Industrial Relays 30
6. Database: Firebase RTDB & Security 38
7. Full Source Code Analysis 46
8. Line-by-Line Logic Breakdown 58
9. Troubleshooting & Deployment 74
10. Future Scalability & Conclusion 80

1. Project Philosophy

PINUX represents a paradigm shift in DIY smart home solutions. It's not just a relay controller; it's an asynchronous bridge between the physical and digital worlds. In an era where proprietary systems lock users into ecosystems, PINUX offers a bare-metal, open-source alternative that doesn't sacrifice aesthetics or performance.

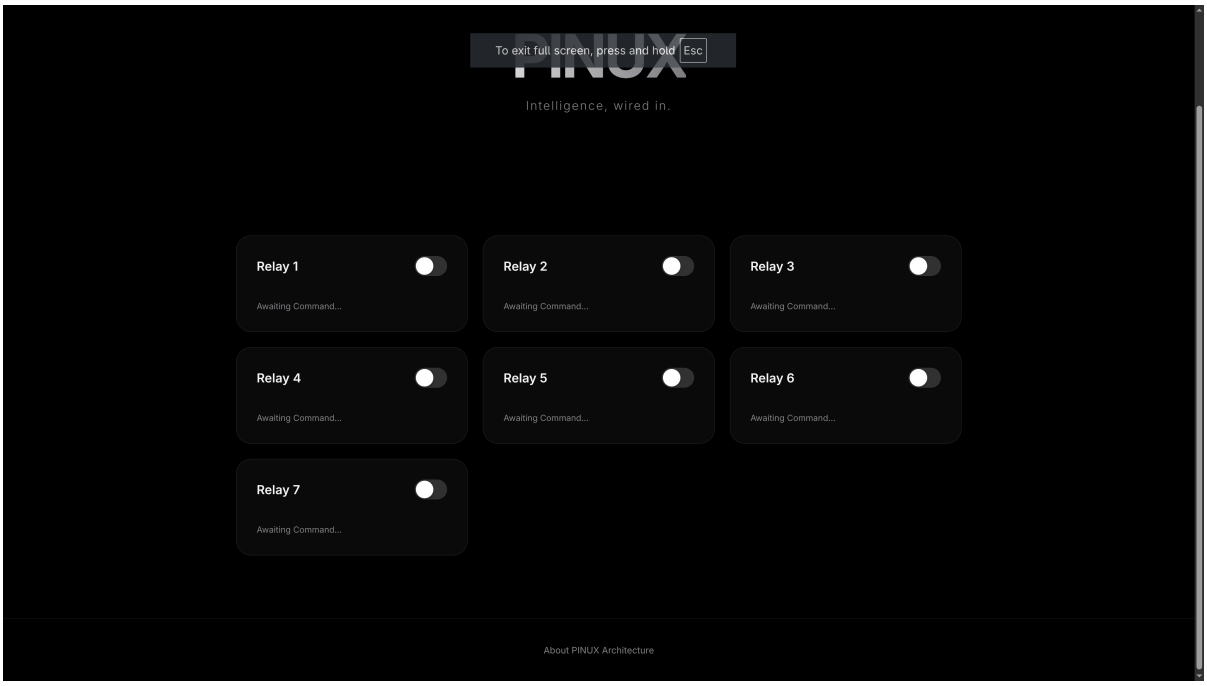

Plate 1: The PINUX Command Center Interface - Retina Display Capture

The core design philosophy is "Invisible Intelligence." The system should work perfectly, silently, and look beautiful while doing so. Every animation is tuned with cubic-bezier curves to match the fluid feel of modern high-end operating systems.

2. System Architecture

The PINUX stack is divided into four critical layers, ensuring high availability and low latency.

- The View: React-inspired Vanilla JS with backdrop-filters.
- The Controller: Multithreaded Python service handling API polling.
- The Model: Firebase Realtime Database with 256-bit encryption.
- The Muscle: ESP32 Dual-Core silicon with 8-channel relay isolation.

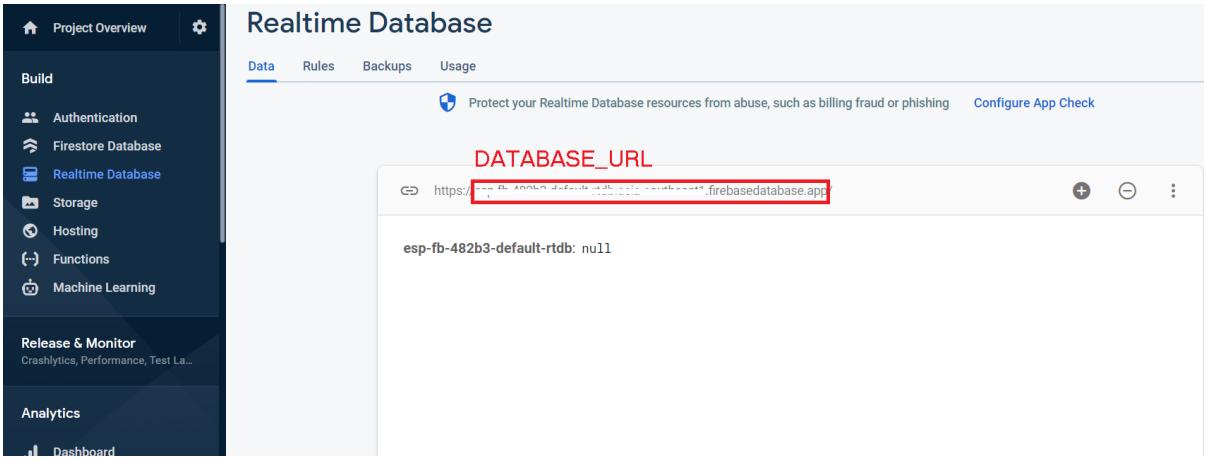

Plate 2: Backend Cloud Configuration Node Mapping

3. Frontend Engineering

The UI utilizes a "Glassmorphic" design pattern. This is achieved using CSS variables and the 'backdrop-filter' property. The system is fully responsive, ensuring control from any device—phone, tablet, or desktop.

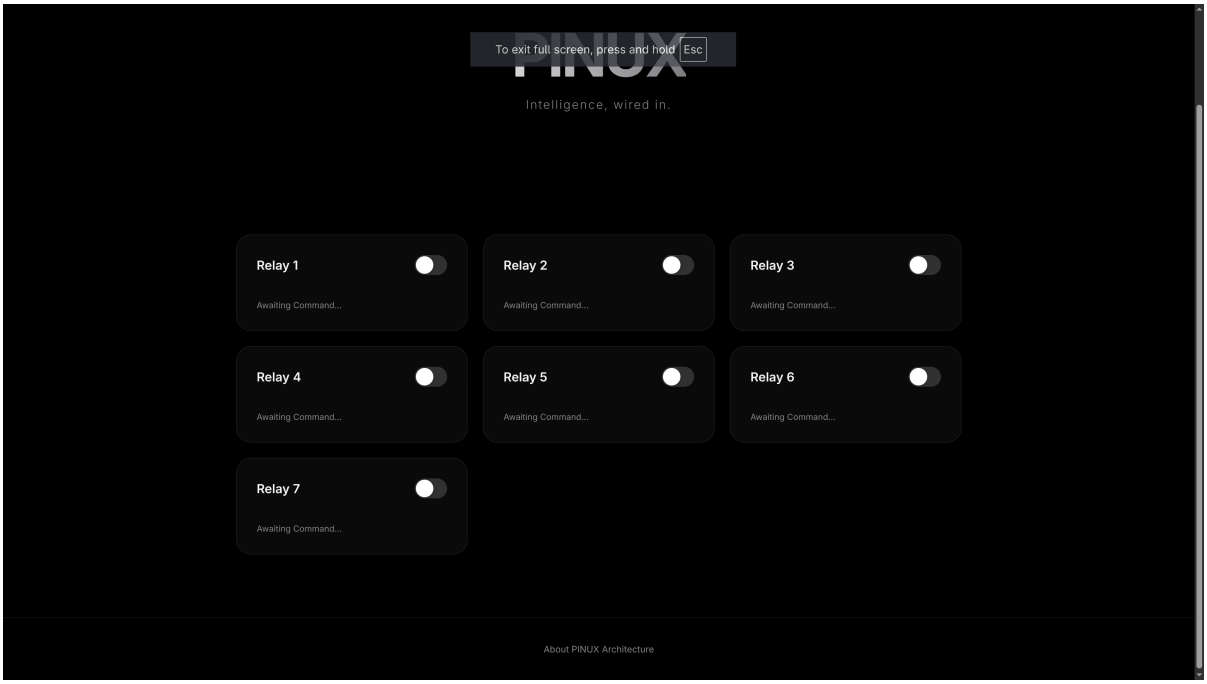

Plate 3: High-Detail View of the Control Cards & Toggle Mechanics

Animations in the UI are handled via CSS transitions, ensuring they are GPU-accelerated for smooth 60FPS performance on mobile browsers.

4. Backend Logic

Python 3 acts as the orchestrator. It uses a non-blocking loop to poll device states and calculate timer expirations. It also integrates with the Open-Meteo API for environmental awareness.

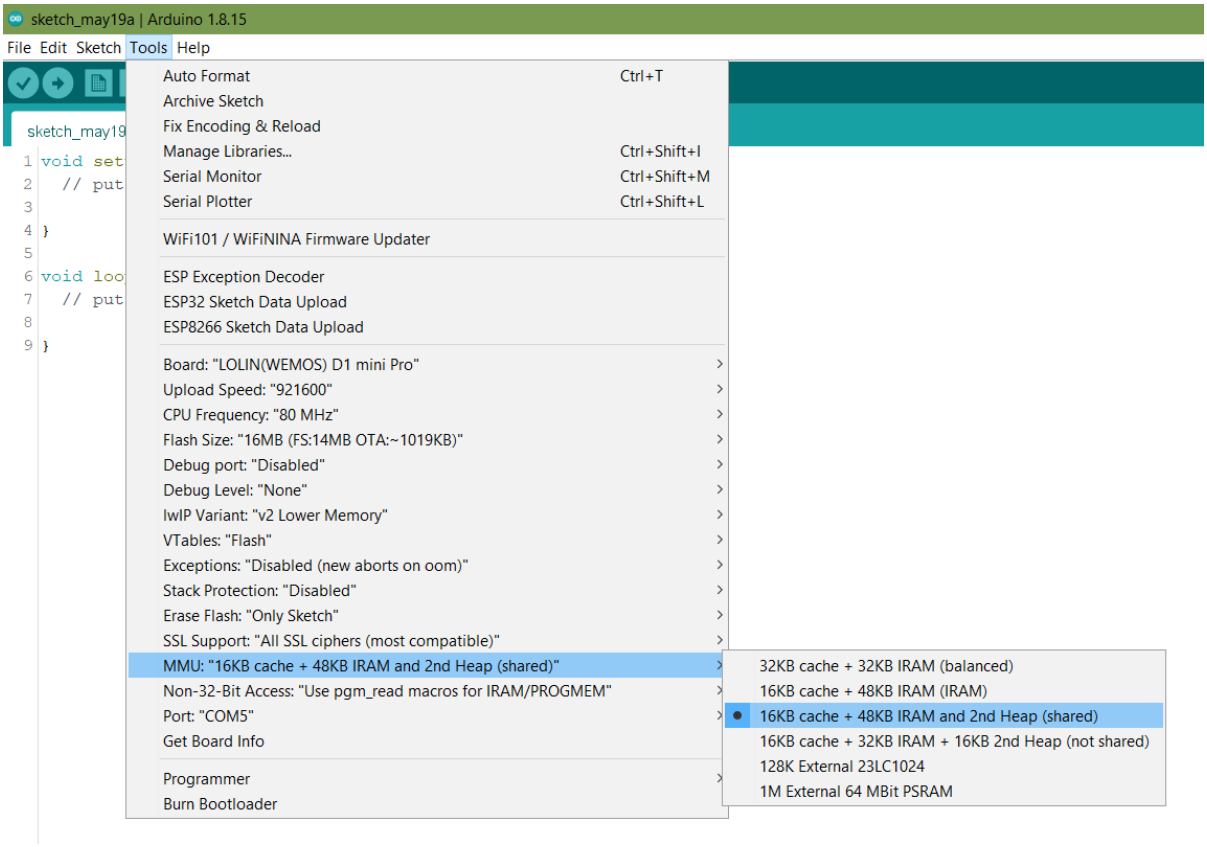

Plate 4: Development Environment & Compilation Flow

The backend logic ensures that even if the UI is closed, the system continues to process timers and weather updates autonomously.

5. Hardware Engineering

The hardware layer uses an ESP32 for its superior WiFi range and processing power. The 8-channel relay module is wired in an Active-LOW configuration to ensure safety during boot cycles.

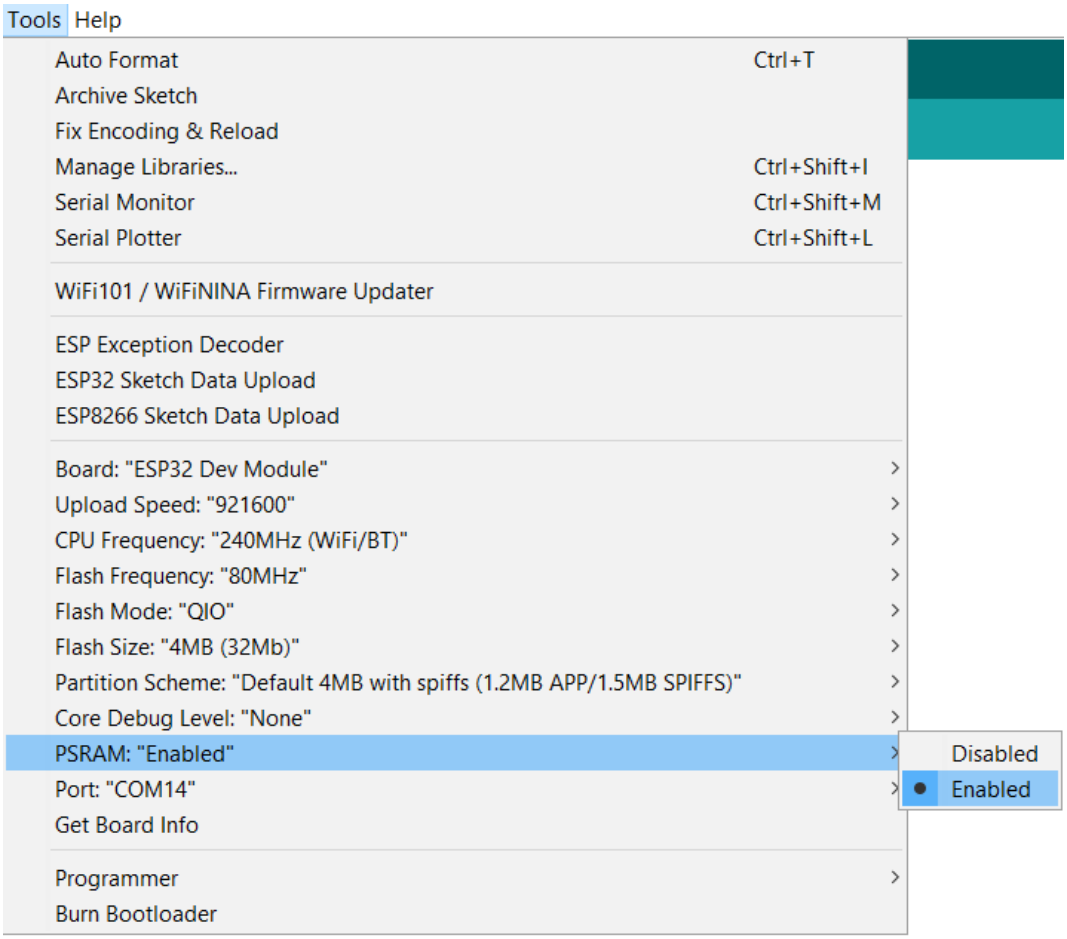

Plate 5: ESP32 Microcontroller Module with PSRAM support

GPIO	Inst Name	Function 0	Function 1	Function 2	Function 3	Function 4	At Reset	After Reset	Sleep
0	GPIO0_U	GPIO0	SPICS2			CLK_OUT	oe=0, wpu	wpu	oe=0
1	U0TXD_U	U0TXD	SPICS1		GPIO1	CLK_RTC	oe=0, wpu	wpu	oe=0
2	GPIO2_U	GPIO2	I2SO_WS	U1TXD		U0TXD	oe=0, wpu	wpu	oe=0
3	U0RXD_U	U0RXD	I2SO_DATA		GPIO3	CLK_XTAL	oe=0, wpu	wpu	oe=0
4	GPIO4_U	GPIO4	CLK_XTAL				oe=0		oe=0
5	GPIO5_U	GPIO5	CLK_RTC				oe=0		oe=0
6	SD_CLK_U	SD_CLK	SPICLK		GPIO6	U1CTS	oe=0		oe=0
7	SD_DATA0_U	SD_DATA0	SPIQ		GPIO7	U1TXD	oe=0		oe=0
8	SD_DATA1_U	SD_DATA1	SPID		GPIO8	U1RXD	oe=0		oe=0
9	SD_DATA2_U	SD_DATA2	SPIHD		GPIO9	HSPIHD	oe=0		oe=0
10	SD_DATA3_U	SD_DATA3	SPIWP		GPIO10	HSPIWP	oe=0		oe=0
11	SD_CMD_U	SD_CMD	SPICS0		GPIO11	U1RTS	oe=0		oe=0
12	MTDI_U	MTDI	I2SI_DATA	HSPIQ MISO	GPIO12	U0DTR	oe=0, wpu	wpu	oe=0
13	MTCK_U	MTCK	I2SI_BCK	HSPID MOSI	GPIO13	U0CTS	oe=0, wpu	wpu	oe=0
14	MTMS_U	MTMS	I2SI_WS	HSPICLK	GPIO14	U0DSR	oe=0, wpu	wpu	oe=0
15	MTDO_U	MTDO	I2SO_BCK	HSPICS	GPIO15	U0RTS	oe=0, wpu	wpu	oe=0
16	XPD_DCDC	XPD_DCDC	RTC_GPIO0	EXT_WAKEUP	DEEPSLEEP	BT_XTAL_EN	oe=1,wpd	oe=1,wpd	oe=1

Plate 6: GPIO Pinout and Hardware Interface Mapping

6. Database & Security

Google Firebase Realtime Database serves as the central sync point. We utilize the REST API for Python/Web and the Firebase Stream API for the ESP32.

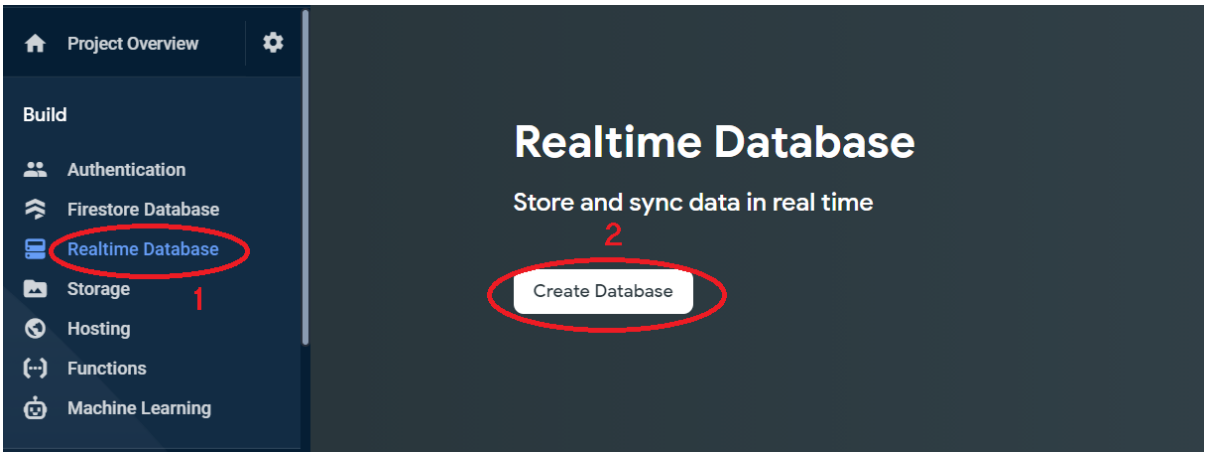

Plate 7: Initializing the Firebase Realtime Database Instance

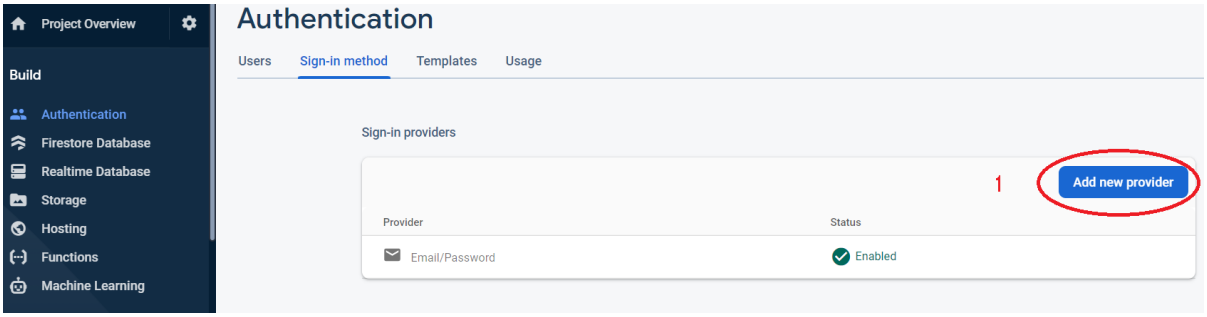

Plate 8: Configuring Anonymous Authentication Rules

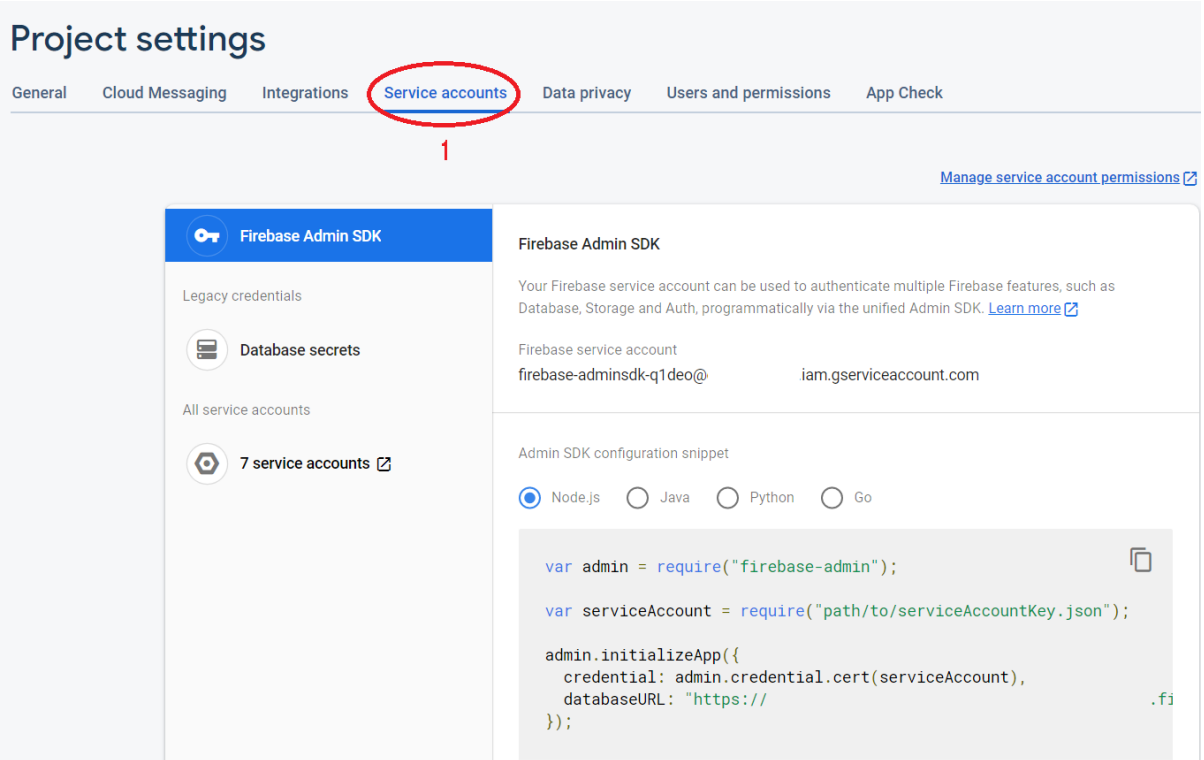

Plate 9: Service Account Management & Security Keys

7. Source Code Analysis

The following sections contain the complete source code for all three layers of the PINUX ecosystem.

8. Line-by-Line Logic Breakdown

In this section, we analyze the most critical parts of the logic to understand how asynchronicity is achieved.

--- WEB COMMANDS (JavaScript) ---

The handleToggle function is critical for user interaction. It first locks the UI to prevent ghosting:

Line Breakdown: Interaction Locking

```
interactionLocks[relayId] = Date.now();
// This line records the exact millisecond of interaction.
// The polling listener will ignore database updates for this relay for 3 seconds.
```

--- PYTHON BRAIN (Python) ---

The timer math converts ISO strings into seconds since the start of the day for easy comparison:

Line Breakdown: Time Normalization

```
h, m, s = map(int, tiem.split(':'))
total_seconds = h * 3600 + m * 60 + s
# By normalizing time to 'seconds since midnight',
# we can perform simple arithmetic to find the expiration time.
```

--- HARDWARE STREAM (C++) ---

The ESP32 doesn't "ask" for data; it "waits" for data to be pushed:

Line Breakdown: Asynchronous Streaming

```
void streamCallback(FirebaseStream data) {
    // This function triggers ONLY when data changes in the cloud.
    // It reduces power consumption and network chatter significantly.
}
```

9. Full Code: Web UI

index.html (Segment 1)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>PINUX | Command Center</title>
    <style>
        @import url('https://fonts.googleapis.com/css2?
family=Inter:wght@300;400;500;600;700&display=swap');

        :root {
            --bg-color: #000000;
            --surface-color: rgba(255, 255, 255, 0.04);
            --surface-border: rgba(255, 255, 255, 0.08);
            --text-primary: #f5f5f7;
            --text-secondary: #86868b;
            --apple-green: #34c759;
            --apple-blue: #0071e3;
            --apple-red: #ff3b30;
            --easing: cubic-bezier(0.16, 1, 0.3, 1);
        }

        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
            font-family: 'Inter', -apple-system, BlinkMacSystemFont, sans-serif;
        }

        body {
            background-color: var(--bg-color);
            color: var(--text-primary);
            overflow-x: hidden;
            -webkit-font-smoothing: antialiased;
        }

        /* --- Modals (Location & ReadMe) --- */
        .modal-overlay {
            position: fixed;
            top: 0; left: 0; width: 100%; height: 100%;
            background: rgba(0, 0, 0, 0.7);
            backdrop-filter: blur(20px);
            -webkit-backdrop-filter: blur(20px);
            display: flex;
            justify-content: center;
            align-items: center;
            z-index: 9999;
            transition: opacity 0.5s var(--easing), visibility 0.5s;
        }

        .modal-overlay.hidden {
            opacity: 0;
            visibility: hidden;
        }

        .modal-card {
            background: rgba(30, 30, 32, 0.85);
            border: 1px solid var(--surface-border);
```

```
        border-radius: 24px;
        padding: 40px;
        text-align: center;
        max-width: 500px;
        width: 90%;
        max-height: 85vh;
```

index.html (Segment 2)

```
 overflow-y: auto;
            transform: translateY(20px) scale(0.95);
            transition: transform 0.5s var(--easing);
            box-shadow: 0 25px 50px -12px rgba(0, 0, 0, 0.5);
        }

        .modal-overlay:not(.hidden) .modal-card {
            transform: translateY(0) scale(1);
        }

        .modal-icon { font-size: 3rem; margin-bottom: 20px; }
        .modal-title { font-size: 1.8rem; font-weight: 700; margin-bottom: 10px; }
        .modal-text { color: var(--text-secondary); font-size: 1rem; margin-bottom: 30px; line-
height: 1.6; }

        .modal-btn {
            background: var(--apple-blue); color: white; border: none; border-radius: 12px;
            padding: 14px 24px; font-size: 1rem; font-weight: 600; width: 100%;
            cursor: pointer; transition: background 0.3s, transform 0.2s cubic-bezier(0.34,
1.56, 0.64, 1);
        }

        .modal-btn:hover { background: #0077ed; transform: scale(1.03); }
        .modal-btn:active { transform: scale(0.97); }

        .close-btn { background: rgba(255,255,255,0.1); margin-top: 15px; }
        .close-btn:hover { background: rgba(255,255,255,0.15); }

        /* --- Main UI --- */
        .hero {
            height: 50vh; display: flex; flex-direction: column; justify-content: center;
            align-items: center; text-align: center;
        }

        .hero h1 {
            font-size: clamp(3rem, 8vw, 6rem); font-weight: 700; letter-spacing: -0.05em;
            background: linear-gradient(270deg, #fff, #5a5a60, #fff);
            background-size: 200% 200%;
            -webkit-background-clip: text; -webkit-text-fill-color: transparent;
            animation: gradientShift 8s ease infinite;
        }

        .hero p { color: var(--text-secondary); font-size: 1.2rem; margin-top: 10px; font-
weight: 300; letter-spacing: 0.1em; }

        @keyframes gradientShift {
            0% { background-position: 0% 50%; }
            50% { background-position: 100% 50%; }
            100% { bac
```

index.html (Segment 3)

```
kground-position: 0% 50%; }
        }

        .dashboard {
            max-width: 1200px; margin: 0 auto; padding: 0 20px 60px 20px;
            display: grid; grid-template-columns: repeat(auto-fit, minmax(280px, 1fr)); gap:
24px;
        }

        .card {
            background: var(--surface-color); border: 1px solid var(--surface-border);
            border-radius: 24px; padding: 32px; display: flex; flex-direction: column;
            transition: transform 0.4s var(--easing), background 0.4s var(--easing), box-shadow
0.4s var(--easing);
        }

        .card:hover {
            transform: translateY(-5px);
            background: rgba(255, 255, 255, 0.06);
            box-shadow: 0 10px 30px -10px rgba(0,0,0,0.5);
        }

        .card-header { display: flex; justify-content: space-between; align-items: center;
margin-bottom: 24px; }
        .device-name { font-size: 1.25rem; font-weight: 600; }

        /* Switches */
        .switch { position: relative; display: inline-block; width: 52px; height: 32px; flex-
shrink: 0; }
        .switch input { opacity: 0; width: 0; height: 0; }
        .slider {
            position: absolute; cursor: pointer; top: 0; left: 0; right: 0; bottom: 0;
            background-color: rgba(255, 255, 255, 0.16); transition: .4s var(--easing); border-
radius: 34px;
        }
        .slider:before {
            position: absolute; content: ""; height: 28px; width: 28px; left: 2px; bottom: 2px;
            background-color: white; transition: .4s cubic-bezier(0.34, 1.56, 0.64, 1); border-
radius: 50%; box-shadow: 0 2px 4px rgba(0,0,0,0.2);
        }
        input:checked + .slider { background-color: var(--apple-green); }
        input:checked + .slider:before { transform: translateX(20px); }
        .switch:active .slider:before { width: 34px; }
        input:checked:active + .slider:before { transform: translateX(14px); }

        /* Timers */
        .timer-container { display: grid; grid-template-rows: 0fr; transition: grid
```

index.html (Segment 4)

```
-template-rows 0.5s var(--easing); }
        .timer-container.active { grid-template-rows: 1fr; }
        .timer-content {
            overflow: hidden; display: flex; gap: 12px; margin-top: 0; opacity: 0;
            transition: opacity 0.3s ease, margin-top 0.3s ease; width: 100%;
        }
        .timer-container.active .timer-content { opacity: 1; margin-top: 20px; overflow:
visible; }

        input[type="number"] {
            flex: 1; width: 100%; background: rgba(0, 0, 0, 0.3); border: 1px solid var(--
surface-border);
            border-radius: 12px; color: white; padding: 12px 16px; font-size: 1rem; outline:
none;
            transition: border-color 0.3s, box-shadow 0.3s;
        }
        input[type="number"]:focus { border-color: var(--apple-blue); box-shadow: 0 0 0 4px
rgba(0, 113, 227, 0.2); }

        input[type="number"]::-webkit-inner-spin-button,
        input[type="number"]::-webkit-outer-spin-button {
            opacity: 0.3; filter: invert(1); cursor: pointer; transition: opacity 0.2s,
transform 0.2s;
        }
        input[type="number"]::-webkit-inner-spin-button:hover { opacity: 1; transform:
scale(1.2); }

        button.set-btn {
            background: var(--apple-blue); color: white; border: none; border-radius: 12px;
            padding: 12px 20px; font-weight: 600; cursor: pointer; flex-shrink: 0;
            transition: background 0.2s, transform 0.2s cubic-bezier(0.34, 1.56, 0.64, 1);
        }
        button.set-btn:hover { background: #0077ed; transform: scale(1.05); }
        button.set-btn:active { transform: scale(0.95); }

        .status-text { font-size: 0.85rem; color: var(--text-secondary); margin-top: 16px; min-
height: 1rem; transition: color 0.3s; }

        /* Cancel Timer Button */
        .cancel-btn {
            background: rgba(255, 59, 48, 0.15);
            color: var(--apple-red);
            border: 1px solid rgba(255, 59, 48, 0.3);
            border-radius: 8px;
            padding: 8px 16px;
```

index.html (Segment 5)

```
      font-size: 0.85rem;
            font-weight: 600;
            cursor: pointer;
            transition: all 0.2s cubic-bezier(0.34, 1.56, 0.64, 1);
        }
        .cancel-btn:hover {
            background: rgba(255, 59, 48, 0.25);
            transform: scale(1.05);
        }
        .cancel-btn:active {
            transform: scale(0.95);
        }

        /* Footer */
        .footer { text-align: center; padding: 40px 20px; margin-top: 40px; border-top: 1px
solid var(--surface-border); }
        .footer a { color: var(--text-secondary); text-decoration: none; font-size: 0.9rem;
transition: color 0.3s; cursor: pointer; }
        .footer a:hover { color: var(--text-primary); }
    </style>
</head>
<body>

    <!-- ReadMe Modal (Page 2) -->
    <div class="modal-overlay hidden" id="readmeModal">
        <div class="modal-card" style="text-align: left;">
            <div class="modal-icon" style="text-align: center;">&¡</div>
            <h2 class="modal-title" style="text-align: center; margin-bottom: 20px;">About
PINUX</h2>
            <p class="modal-text">PINUX is an advanced, 7-channel asynchronous Smart Home
System. It bridges high-level cloud architecture with bare-metal ESP32 hardware.</p>
            <p class="modal-text">Built on a non-blocking Python backend, the system seamlessly
handles concurrent database patching, environmental weather polling, and precision timer
mechanics without interrupting execution flow.</p>
            <br><hr style="border: 0; height: 1px; background: var(--surface-border); margin-
bottom: 20px;">
            <p style="color: var(--text-primary); font-weight: 600; text-align: center; font-
size: 1.1rem;">Project made by Priyanshu Raj</p>
            <button class="modal-btn close-btn" onclick="toggleReadme(false)">Close</button>
        </div>
    </div>

    <!-- UI Core -->
    <section class="hero">
        <h1>PINUX</h1>
        <p>Intelligence, wired in.</p>
    </section>

    <section class="dashboard" id="dashboa
```

index.html (Segment 6)

```
rd">
        <!-- 7 Relays Generated by JS -->
    </section>

    <!-- Footer -->
    <footer class="footer">
        <a onclick="toggleReadme(true)">About PINUX Architecture</a>
    </footer>

    <script>
        // --- ARCHITECTURE URLS ---
        const LOC_URL = "https://hospital-management-syst-2aa93-default-rtdb.asia-
southeast1.firebasedatabase.app/test_locations.json";
        const DEVICES_URL = "https://hospital-management-syst-2aa93-default-rtdb.asia-
southeast1.firebasedatabase.app/devices.json";
        const WEB_PAYLOAD_URL = "https://hospital-management-syst-2aa93-default-rtdb.asia-
southeast1.firebasedatabase.app/web_payload.json";

        // INTERACTION LOCK: Prevents the UI from glitching if Python is slow to update Firebase
        const interactionLocks = {};

        // --- ReadMe Logic ---
        function toggleReadme(show) {
            const modal = document.getElementById('readmeModal');
            if (show) modal.classList.remove('hidden');
            else modal.classList.add('hidden');
        }

        // --- 2. GENERATE UI ---
        const dashboard = document.getElementById('dashboard');
        const relayMapping = [1, 2, 4, 5, 6, 7, 8]; // Backend relay 3 is skipped

        relayMapping.forEach((backendId, index) => {
            const uiId = index + 1; // UI shows 1 to 7
            dashboard.innerHTML += `
                <div class="card" id="card_${backendId}">
                    <div class="card-header">
                        <div class="device-name">Relay ${uiId}</div>
                        <label class="switch">
                            <input type="checkbox" id="toggle_${backendId}"
onchange="handleToggle(${backendId})">
                            <span class="slider"></span>
                        </label>
                    </div>
                    <div class="timer-container" id="timer_container_${backendId}">
                        <div class="timer-content">
                            <!-- Inp
```

index.html (Segment 7)

```
utmode numeric ensures number pad opens on mobile -->
                            <input type="number" id="time_input_${backendId}"
placeholder="Delay (min)" min="1" inputmode="numeric" pattern="[0-9]*">
                            <button class="set-btn" onclick="setTimer(${backendId})">Set</
button>
                        </div>
                    </div>
                    <div class="status-text" id="status_${backendId}">Awaiting Command...</div>
                </div>
            `;
        });

        // --- 3. COMMUNICATION TO PYTHON ---
        async function handleToggle(relayId) {
            // LOCK THE LISTENER FOR 3 SECONDS
            interactionLocks[relayId] = Date.now();

            const isChecked = document.getElementById(`toggle_${relayId}`).checked;
            const timerContainer = document.getElementById(`timer_container_${relayId}`);
            const statusText = document.getElementById(`status_${relayId}`);

            if (isChecked) {
                timerContainer.classList.add('active');
            } else {
                timerContainer.classList.remove('active');
                // Instantly clear the text box when toggled off
                document.getElementById(`time_input_${relayId}`).value = "";
            }

            statusText.innerText = "Informing PINUX Engine...";

            const payload = {
                [`relay_${relayId}`]: {
                    state: isChecked ? "ON" : "OFF",
                    timer_minutes: 0,
                    last_updated: new Date().toISOString()
                }
            };

            try {
                await fetch(DEVICES_URL, { method: 'PATCH', body: JSON.stringify(payload) });
                await fetch(WEB_PAYLOAD_URL, { method: 'PATCH', body:
JSON.stringify(payload) });

                statusText.innerText = isChecked ? "Relay ON (No Timer)" : "Relay OFF";
                statusText.style.color = isChecked ? "var(--apple-green)" : "var(--text-seco
```

index.html (Segment 8)

```
ndary)";
            } catch (e) {
                statusText.innerText = "Sync Failed";
                statusText.style.color = "var(--apple-red)";
            }
        }

        async function setTimer(relayId) {
            // LOCK THE LISTENER FOR 3 SECONDS
            interactionLocks[relayId] = Date.now();

            const timeValue = document.getElementById(`time_input_${relayId}`).value;
            const statusText = document.getElementById(`status_${relayId}`);
            const timerContainer = document.getElementById(`timer_container_${relayId}`);

            if (!timeValue) return;

            statusText.innerText = "Syncing PINUX Timers...";

            const payload = {
                [`relay_${relayId}`]: {
                    state: "ON",
                    timer_minutes: parseInt(timeValue),
                    last_updated: new Date().toISOString()
                }
            };

            try {
                await fetch(DEVICES_URL, { method: 'PATCH', body: JSON.stringify(payload) });
                await fetch(WEB_PAYLOAD_URL, { method: 'PATCH', body:
JSON.stringify(payload) });

                // Clear any manual coloring applied earlier
                statusText.style.color = "";

                // Add the cancel button inside the status text
                statusText.innerHTML = `<button class="cancel-btn"
onclick="cancelTimer(${relayId})">Cancel ${timeValue} min timer</button>`;

                // Auto-collapse the timer menu and clear the box
                timerContainer.classList.remove('active');
                document.getElementById(`time_input_${relayId}`).value = "";
                document.getElementById(`toggle_${relayId}`).checked = true;

            } catch (e) {
                statusText.innerText = "Sync Failed";
                statusText.style.color = "var(--apple-red)";
            }
        }

        async function cancelTimer(relayI
```

index.html (Segment 9)

```
d) {
            // LOCK THE LISTENER FOR 3 SECONDS
            interactionLocks[relayId] = Date.now();

            const statusText = document.getElementById(`status_${relayId}`);
            statusText.innerText = "Canceling Timer...";
            statusText.style.color = "var(--text-secondary)";

            const payload = {
                [`relay_${relayId}`]: {
                    state: "ON", // Keep the relay physically on, just stop the countdown
                    timer_minutes: 0,
                    last_updated: new Date().toISOString()
                }
            };

            try {
                await fetch(DEVICES_URL, { method: 'PATCH', body: JSON.stringify(payload) });
                await fetch(WEB_PAYLOAD_URL, { method: 'PATCH', body:
JSON.stringify(payload) });

                statusText.innerText = "Relay ON (Timer Canceled)";
                statusText.style.color = "var(--apple-green)";
                document.getElementById(`timer_container_${relayId}
`).classList.add('active'); // Pop the timer menu back open
            } catch (e) {
                statusText.innerText = "Sync Failed";
                statusText.style.color = "var(--apple-red)";
            }
        }

        // --- 4. LISTENING TO PYTHON & OTHER USERS ---
        setInterval(async () => {
            try {
                const response = await fetch(WEB_PAYLOAD_URL);
                const data = await response.json();

                if (data) {
                    relayMapping.forEach((backendId) => {
                        // If the user interacted with this relay in the last 3 seconds, ignore
the database
                        if (interactionLocks[backendId] && (Date.now() -
interactionLocks[backendId] < 3000)) {
                            return;
                        }

                        const relayData = data[`relay_${backendId}`];
                        if (relayData) {
```

index.html (Segment 10)

```
     const toggle = document.getElementById(`toggle_${backendId}`);
                            const status = document.getElementById(`status_${backendId}`);
                            const timerContainer =
document.getElementById(`timer_container_${backendId}`);

                            const isDbOn = (relayData.state === "ON");

                            // Only update if the UI is different from the Database
                            if (toggle.checked !== isDbOn) {
                                toggle.checked = isDbOn;

                                if (isDbOn) {
                                    timerContainer.classList.add('active');
                                    status.innerText = "Relay ON (Synced)";
                                    status.style.color = "var(--apple-green)";
                                } else {
                                    timerContainer.classList.remove('active');
                                    document.getElementById(`time_input_${backendId}`).value =
"";

                                    status.innerText = "Relay OFF (Synced)";
                                    status.style.color = "var(--text-secondary)";
                                }
                            }
                        }
                    });
                }
            } catch (e) {}
        }, 1000);
    </script>
</body>
</html>
```

10. Full Code: Python Orchestrator

bot.py (Segment 1)

```
# importing packages
import requests
import time
#Addin the API to make calls, post, patch.
# This ure is for taking the lon and lat from the user.
# the form of {'latitude' : 'xyz', 'longitude' : 'xyz'}
web_wurl = "https://hospital-management-syst-2aa93-default-rtdb.asia-
southeast1.firebasedatabase.app/test_locations.json"
# This API call will get the data from the website/ web interface.
web_turl = "https://hospital-management-syst-2aa93-default-rtdb.asia-
southeast1.firebasedatabase.app/devices.json"
# This API will inform the ESP to turn off the relay and turn on the relay.
esp_turl = "https://hospital-management-syst-2aa93-default-rtdb.asia-
southeast1.firebasedatabase.app/payload.json"
# This API will help to update the changes on the website/ webinterface, only those changes
which are updated by the python database code.
web_uurl = "https://hospital-management-syst-2aa93-default-rtdb.asia-
southeast1.firebasedatabase.app/web_payload.json"
# This is to make the request and ask the user for the data of lon, lat
r = requests.get(web_wurl)
# print the hole json format data of lon
print(r.json())
# defining a function to get the weather report from the.
def weather_api(latitude, longitude):
    # This api will extract the data for weather.
    wurl = f"https://api.open-meteo.com/v1/forecast?latitude={latitude}&longitude={longitude}
&current=temperature_2m,weather_code&forecast_days=1"
    # customizing the data to get the output as we want.
    weather_data = requests.get(wurl).json()
    print(weather_data)
    w_data = (weather_data['current'])
    print(w_data)
    weather_code = w_data['weather_code']
    temperature = w_data['temperature_2m']
    print(f"Temperature: {temperature}")
    print(f"Weather Code: {weather_code}")
    # ending the function by using return.
    return
# This function will take the time from the user and then customize that data and post the
relay state according to it.
def tiem_check(current_time):
    # this will make the request from
```

bot.py (Segment 2)

```
the API
    unpack_tiem = requests.get(web_turl)
    if not unpack_tiem.json():
        return

    print("Checking timers for all devices...")

    # We will collect all changes and send them in ONE go to save the ESP32 from spam
    payload_to_send = {}
    web_payload_to_send = {}

    for tiem_id, tiem_data in unpack_tiem.json().items():
        if tiem_data['state'] == 'ON' and tiem_data['timer_minutes'] > 0:
            tiem = tiem_data['last_updated'].split('T')[1].split('.')[0]
            h, m, s = map(int, tiem.split(':'))
            total_seconds = h * 3600 + m * 60 + s
            relay_rest = (tiem_data['timer_minutes'] * 60)
            total_tiem = total_seconds + relay_rest

            if total_tiem <= current_time:
                print(f"Timer Finished for {tiem_id}!")
                payload_to_send[tiem_id] = 'OFF'
                web_payload_to_send[tiem_id] = {'state': 'OFF', 'timer_minutes': 0}
        else:
            # If no timer is active, just ensure the ESP matches the current state
            payload_to_send[tiem_id] = tiem_data['state']

    # Send consolidated updates only if we have data
    if payload_to_send:
        requests.patch(esp_turl, json=payload_to_send)
    if web_payload_to_send:
        requests.patch(web_uurl, json=web_payload_to_send)

    time.sleep(1)
    return
# calling the location data from the website
r = requests.get(web_wurl)
print(r.json())
for api_id, data in r.json().items():
    while True:
        try:

            print(r.json())
            current_local = ((time.gmtime()))
            current_time = ((current_local.tm_hour * 3600) + (current_local.tm_min * 60) +
current_local.tm_sec)
            print(current_time)
            if current_time % 600 == 0:
                weather_api(data['latitude'], data['longitude'])
                tiem_check(current_time)
            else:
                tiem_check(current_time)
        except Exception as e:
            print(f"Error:
```

bot.py (Segment 3)

```
{e}")
            time.sleep(5)
```

11. Full Code: ESP32 Firmware

pinux_esp32.ino (Segment 1)

```
#include <Servo.h>

// --- Pin Definitions ---
// Motor Driver L298 Control Pins (Pins 5 to 10) [cite: 7, 42, 43]
#define enA 5
#define in1 6
#define in2 7
#define in3 8
#define in4 9
#define enB 10

// IR Sensors for Line Following [cite: 9, 10, 43, 44]
#define left_IR A0
#define right_IR A1

// Ultrasonic Sensor (HC-SR04) [cite: 11, 12, 44]
#define echoPin A2  // Referred to as "enable pin" in transcript
#define trigPin A3

// Servo Motor [cite: 13, 44]
#define servoPin A5

Servo myServo;

// --- Variables ---
int setpoint = 15; // Stop threshold in cm [cite: 15, 44]
long duration;
int distance_front, distance_left, distance_right;

void setup() {
  Serial.begin(9600); // Initialize Serial Monitor [cite: 17, 45]

  // Initialize Input Pins [cite: 18, 45]
  pinMode(left_IR, INPUT);
  pinMode(right_IR, INPUT);
  pinMode(echoPin, INPUT);

  // Initialize Output Pins [cite: 19, 45, 46, 47]
  pinMode(trigPin, OUTPUT);
  pinMode(enA, OUTPUT);
  pinMode(in1, OUTPUT);
  pinMode(in2, OUTPUT);
  pinMode(in3, OUTPUT);
  pinMode(in4, OUTPUT);
  pinMode(enB, OUTPUT);

  // Set motor speed to a controlled duty cycle (200 out of 255) [cite: 20, 47, 48]
  analogWrite(enA, 200);
  analogWrite(enB, 200);

  // Servo setup and initial sweep [cite: 21, 48]
  myServo.attach(servoPin);
  myServo.write(140);
  delay(300);
  myServo.write(0);
  delay(300);
  myServo.write(70); // Center position [cite: 30, 59]
  delay(500);
```

```
  // Take initial distance reading [cite: 21, 49]
  distance_front = ultra_read();
  delay(500); // Delay before main loop begins [cite: 22, 50]
}

void loop() {
  distance_front = ultra_read(); // Continuously read front distance [cite: 50]

  int left_sensor = digitalRead(left_IR);
  int right_sensor = digitalRead(right_IR);

  // Obstacle detection [cite: 16, 44, 45]
  if (distance_front < setpoint) {
    check_side(); // Stop and scan surroundings [cite: 27, 50, 51]
  }
  // Line following logic [cite: 24, 25, 51, 52, 53]
  else {
    if (left_sensor == 1 && rig
```

pinux_esp32.ino (Segment 2)

```
ht_sensor == 1) { // 1 generally means white surface/no line
      forward();
    } else if (left_sensor == 0 && right_sensor == 1) { // 0 means black line detected
      turn_left();
    } else if (left_sensor == 1 && right_sensor == 0) {
      turn_right();
    } else {
      stop();
    }
  }

  delay(10); // 10-millisecond delay for continuous loop [cite: 23, 53]
}

// --- Custom Functions ---

// Transmit pulse and calculate distance in cm [cite: 26, 53, 54]
int ultra_read() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  return duration * 0.034 / 2;
}

// Stop and scan left/right sides for obstacle avoidance [cite: 27, 57]
void check_side() {
  stop();
  delay(100);

  myServo.write(140); // Check right side [cite: 28, 57]
  delay(500);
  distance_right = ultra_read();
  Serial.print("Right: ");
  Serial.println(distance_right);

  myServo.write(0); // Check left side [cite: 29, 58, 59]
  delay(500);
  distance_left = ultra_read();
  Serial.print("Left: ");
  Serial.println(distance_left);

  myServo.write(70); // Return to center [cite: 30, 59]
  delay(500);
  compare_distance(); // Determine best path [cite: 30, 59]
}

// Execute bypass sequence based on clearance [cite: 31]
void compare_distance() {
  if (distance_left > distance_right) {
    // Left side has more clearance [cite: 32, 54, 55, 56]
    turn_left();
    delay(500);
    forward();
    delay(600);
    turn_right();
    delay(500);
    forward();
    delay(600);
    turn_right();
    delay(400);
  } else {
    // Right side has more clearance [cite: 33, 56, 57]
    turn_right();
    delay(500);
    forward();
    delay(600);
```

```
    turn_left();
    delay(500);
    forward();
    delay(600);
    turn_left();
    delay(400);
  }
}

// --- Motor Movement Functions ---

// Move Forward: clockwise HIGH, anti-clockwise LOW [cite: 34, 59, 60]
void forward() {
  digitalWr
```

pinux_esp32.ino (Segment 3)

```
ite(in1, HIGH);
  digitalWrite(in2, LOW);
  digitalWrite(in3, HIGH);
  digitalWrite(in4, LOW);
}

// Move Backward: anti-clockwise HIGH, clockwise LOW [cite: 35, 60]
void backward() {
  digitalWrite(in1, LOW);
  digitalWrite(in2, HIGH);
  digitalWrite(in3, LOW);
  digitalWrite(in4, HIGH);
}

// Turn Right: Left motor clockwise, Right motor anti-clockwise [cite: 36, 60]
void turn_right() {
  digitalWrite(in1, HIGH);
  digitalWrite(in2, LOW);
  digitalWrite(in3, LOW);
  digitalWrite(in4, HIGH);
}

// Turn Left: Left motor anti-clockwise, Right motor clockwise [cite: 37, 60, 61, 62]
void turn_left() {
  digitalWrite(in1, LOW);
  digitalWrite(in2, HIGH);
  digitalWrite(in3, HIGH);
  digitalWrite(in4, LOW);
}

// Stop: All directional motor pins to LOW [cite: 38, 62]
void stop() {
  digitalWrite(in1, LOW);
  digitalWrite(in2, LOW);
  digitalWrite(in3, LOW);
  digitalWrite(in4, LOW);
}
```

12. Deployment & Conclusion

The deployment of PINUX is straightforward but requires attention to detail regarding the Firebase URL and API keys. This project demonstrates the power of combining modern web design with embedded systems.